SSC MTS 2024 question paper with answers

21 October 2024 · Shift 3

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.3

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 3% per year rate of interest for 3 years.

(A) 230
(B) 310
(C) 210
(D) 270

Q.4

Find the LCM of 8, 12, 18, 24 and 36.

(A) 27
(B) 36
(C) 72
(D) 64

Q.5

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 3% per year rate of interest for 4 years.

(A) 300
(B) 240
(C) 340
(D) 260

Q.6

Maria buys a dress for ₹3,000. She could not decide between a discount of 35% and three successive discounts of 20%, 10% and 5%. The difference between both discounts (in ₹) is:

(A) 0
(B) 102
(C) 20
(D) 68

Q.7

Geeta went to her home town at a distance of 270 km from her house. She started her journey at 10:00 a.m. and reached the destination at 02:00 p.m. by travelling 20 km by cab at a speed of 60 km/h and 240 km by train running at a speed of 80 km/h. She covered the remaining distance by an auto to reach her place. What was the speed of the auto?

(A) 15 km/h
(B) 20 km/h
(C) 18 km/h
(D) 4 km/h

Q.8

Shikha sells a bookshelf to Vansh at a loss of 5%, Vansh sells the same bookshelf to Swarnima at a profit of 10%, and Swarnima sells the same bookshelf to Nikhil at a loss of 20%. If Nikhil purchased the bookshelf for ₹16,720, then what was the cost (in ₹) of the bookshelf for Shikha?

(A) 18,000
(B) 21,500
(C) 19,500
(D) 20,000

Q.10

The average age of a husband and wife, and their child 3 years ago, was 27 years, and that of the wife and the child 5 years ago was 20 years. The present age of the husband is:

(A) 45 years
(B) 35 years
(C) 40 years
(D) 50 years

Q.11

Rohit made a profit of 22% when he sold a jacket for ₹7,930. Find the cost price of the jacket.

(A) ₹7,500
(B) ₹7,000
(C) ₹6,000
(D) ₹6,500

Q.13

15 kg of superior quality of wheat is mixed with 5 kg of inferior quality of wheat. The price of superior quality and inferior quality wheat is ₹80 and ₹60, respectively. Find the average price per kg of the mixture.

(A) ₹70
(B) ₹65
(C) ₹150
(D) ₹75

Q.15

The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher.

(A) 24 years
(B) 26 years
(C) 27 years
(D) 25 years

Q.16

The average age of 40 students in a class is 12 years. The average age of the students increased by 4 months due to 20 new admissions in the class. The average age of the newly admitted students is:

(A) 14 years
(B) 13 years
(C) 14 years 6 months
(D) 13 years 4 months

Q.17

What is the single equivalent price increase for two successive price increases of 12% and 18% on an article?

(A) 15%
(B) 32.16%
(C) 30.24%
(D) 30%

Q.18

On a 200-m-long circular track, two people start from the same point, at the same time, at speeds of 12 km/h and 15 km/h, respectively. After how much time will they meet again for the first time if they are running in the same direction?

(A) 220 seconds
(B) 200 seconds
(C) 240 seconds
(D) 180 seconds

Q.19

The sum of length and breadth of a rectangle is 68 m and the ratio of length and breadth is 11:23. What is the area of the rectangle (in m^2)?

(A) 1012
(B) 976
(C) 253
(D) 1331

Q.20

If p is directly proportional to q and is p = 7 when q = 6, then what is the value of p when q = 21?

(A) 24.5
(B) 25.0
(C) 24.0
(D) 23.5

Reasoning Ability and Problem Solving 15 questions

Q.1

BEAR is related to YBXO in a certain way based on the English alphabetical order. In the same way, WATCH is related to TXQZE. To which of the following is LOTUS related, following the same logic?

(A) IKQRP
(B) IJQQP
(C) IJRRP
(D) ILQRP

Q.2

What will come in the place of '?' in the following equation, if '–' and '×' are interchanged and '+' and '÷' are interchanged? 28 ÷ 18 + 6 × 6 – 4 = ?

(A) 17
(B) 7
(C) 14
(D) 9

Q.3

What should come in place of the question mark (?) in the given series? 25000, 5000, 1000, 200, 40, ?

(A) 16
(B) 20
(C) 12
(D) 8

Q.4

12 is related to 61 following a certain logic. Following the same logic, 15 is related to 76. To which of the following is 16 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 81
(B) 86
(C) 84
(D) 88

Q.7

In a certain code language, 'they are smart' is written as 'bt mt rt' and 'she is smart' is written as 'rt kt st'. How is 'smart' written in the given code language?

(A) mt
(B) rt
(C) st
(D) kt

Q.8

What should come in place of the question mark (?) in the given series based on the English alphabetical order? GKS, MQY, SWE, YCK, ?

(A) EIO
(B) EWR
(C) EIQ
(D) EQW

Q.9

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (12, 77, 5) (21, 87, 3)

(A) (15, 127, 7)
(B) (17, 401, 9)
(C) (12, 197, 3)
(D) (11, 188, 6)

Q.12

Seven people R, S, T, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). G is sitting second to the right of E. Only one person is sitting between G and R. Only one person is sitting between R and T. F is sitting second to the left of D. D is an immediate neighbour of T. How many people are sitting between S and T when counted from the left of S?

(A) One
(B) Two
(C) Four
(D) Three

Q.13

Which two signs should be interchanged to make the given equation correct? $45 - 33 \div 3 \times 4 + 67 = 22$

(A) – and ×
(B) – and +
(C) ÷ and ×
(D) + and ×

Q.14

In a certain code language, 'JOKE' is coded as 'NIOY' and 'TRICKY' is coded as 'XLMWOS'. What is the code for 'SPEAKERS' in the given code language?

(A) WJHUOYVM
(B) WJIUOYUM
(C) WJILOYVM
(D) WJIUOYVM

Q.15

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All crocs are frogs. All crocs are turtles. Conclusions: (I) All turtles are frogs. (II) At least some frogs are crocs.

(A) Only conclusion (II) follows.
(B) Only conclusion (I) follows.
(C) Neither conclusion (I) nor (II) follows.
(D) Both conclusions (I) and (II) follow.

Q.16

In a certain code language, 253 stands for 'chairs are old', 378 stands for 'make good chairs' and 546 stands for 'lady is old'. Which of the following stands for 'are' in the same code language?

(A) 4
(B) 2
(C) 5
(D) 3

Q.17

Six people A, B, C, D, E and F are sitting around a circular table, facing the centre (but not necessarily in the same order). E is sitting second to the left of C. C is an immediate neighbour of both A and B. F is sitting third to the right of A. How many people are sitting between D and C when counted from the left of D?

(A) 4
(B) 1
(C) 3
(D) 2

Q.18

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All bongos are sitars. No cymbal is a sitar. Conclusions: [1] All sitars are bongos. [2] Some bongos are cymbals.

(A) Neither conclusion [1] nor [2] follows
(B) Only conclusion [2] follows
(C) Only conclusion [1] follows
(D) Both conclusions [1] and [2] follow

Q.20

QFRG is related to NCVK in a certain way based on the English alphabetical order. In the same way, KZZO is related to HWDS. To which of the following is ETHW related, following the same logic?

(A) APLZ
(B) BPMZ
(C) CRKB
(D) BQLA

General Awareness 25 questions

Q.1

In India, documented by a study, in which level of education is the majority of the population unemployed?

(A) Primary school
(B) Middle school
(C) Graduate and post-graduate
(D) Matriculates

Q.2

In June 2023, which state Government launched a project named 'Pride' to provide employment opportunities to the transgender community in diverse and innovative sectors?

(A) Goa
(B) Kerala
(C) Rajasthan
(D) Sikkim

Q.3

In the South Indian temples, the pyramidal tower is called the _____.

(A) shikhara
(B) mandapa
(C) vimana
(D) vahan

Q.4

What is the main purpose of the office of the Comptroller and Auditor General (CAG)?

(A) Investigating and reporting on financial transactions
(B) Regulating tax collection
(C) Formulating the national budget
(D) Enforcing government policies

Q.5

Coimbatore, which is located in the state of _____, is famous for the cotton textile industry.

(A) Kerala
(B) Tamil Nadu
(C) Andhra Pradesh
(D) Karnataka

Q.6

Which of the following States/Union Territories is included in the 'Areas of Inadequate Rainfall' of India?

(A) Haryana
(B) Andaman and Nicobar Islands
(C) Ladakh
(D) Meghalaya

Q.7

Hampi Utsava is a colourful cultural festival that emulates the ethos and grandeur of which of the following empires?

(A) Kadamba
(B) Vijayanagara
(C) Chalukya
(D) Satavahana

Q.8

Which of the following are edible polysaccharides?

(A) Methyl cellulose, chitin and actin
(B) Sucrose, glucose and cellulose
(C) Cellulose, chitin and myosin
(D) Starch, alginates, pectins and chitosans

Q.9
In December 2023, which former Tata Consultancy Services (TCS) CEO and MD joined the Indian Institute of Technology Bombay (IIT Bombay) as Head of Translational Research and Entrepreneurship?

(A) Natarajan Chandrasekaran
(B) Rajesh Gopinathan
(C) FC Kohli
(D) Subramaniam Ramadorai

Q.10
Which of the following is a molecular compound?

(A) NaCl
(B) $MgSO_4$
(C) CO_2
(D) CaO

Q.11
Eutrophication is a condition where excessive growth of algal bloom occurs on the surface of a water body because of abundant nutrients. Which of the following are the key nutrients for eutrophication?

(A) Iron and nitrogen
(B) Phosphorus and nitrogen
(C) Nitrogen and potassium
(D) Phosphorus and iron

Q.12
Mangalampalli Balamuralikrishna was an Indian Carnatic singer born in Sankaraguptam village of which state?

(A) Kerala
(B) Tamil Nadu
(C) Karnataka
(D) Andhra Pradesh

Q.13
Who among the following is the author of the book 'The Hungry Tide'?

(A) Mulk Raj Anand
(B) Ruskin Bond
(C) RK Narayan
(D) Amitav Ghosh

Q.14
The Parliamentary form of Government is a key feature given in the Indian Constitution and its provisions are detailed in Articles _____ of Part V (Chapter II).

(A) 79 to 122
(B) 170 to 203
(C) 124 to 168
(D) 211 to 236

Q.15
Which of the following countries hosted the ACC Under-19 Asia Cup cricket in the year 2023?

(A) Pakistan
(B) Sri Lanka
(C) India
(D) UAE

Q.16
Who among the following was the first President of Indian National congress?

(A) George Yule
(B) Womesh Chandra Banerjee
(C) Badruddin Tyabji
(D) Dadabhai Naoroji

Q.17
Which of the following raw materials is NOT used to prepare the baking soda?

(A) Na_2CO_3
(B) NaCl
(C) NH_3
(D) H_2O

Q.18
In the year 2010, Commonwealth Games was hosted by which of the following countries?

(A) Canada
(B) England
(C) Australia
(D) India

Q.19
'Dollu Kunitha' is primarily a folk dance of which of the following Indian states?

(A) Goa
(B) Kerala
(C) Tamil Nadu
(D) Karnataka

Q.20
What is the duration for which a member of a House of Legislature of a State is, without permission of the House, absent from all meetings, thereof the house may declare his seat vacant?

(A) 180 days
(B) 60 days
(C) 90 days
(D) 30 days

Q.21
Odissi dancer _____ was awarded the Sangeet Natak Akademi Award 2022 in the dance category.

(A) Kumkum Mohanty
(B) Shubhada Varadkar
(C) Niranjan Rout
(D) Debi Vasu

Q.22
Who among the following was the famous ruler of the Gauda Kingdom?

(A) Shashanka
(B) Grahavarman
(C) Rajyavardhan
(D) Rudradaman

Q.23
Select the INCORRECT statement from the following.

(A) The first cotton textile mill in India was established in Bombay in 1854.
(B) The development of the cotton textile industry in Britain led to the decline of Indian textile producers.
(C) Cotton clothes made in England had driven Indian clothes out of the markets of Europe and America in the beginning of the 20th century.
(D) The Sale and Devang communities of South India were famous for weaving.

Q.24
According to the census of India 2011, which of the following states has the highest percentage of rural population?

(A) Assam
(B) Uttar Pradesh
(C) West Bengal
(D) Bihar

Q.25
The famous Jain temple of Dilwara, built by Solanki rulers, is located at which place?

(A) Mount Abu
(B) Ajmer
(C) Surat
(D) Jaipur

English Language and Comprehension 23 questions

Q.2
Select the most appropriate ANTONYM for the underlined word in the given sentence. Such specifications have revived our designers and manufacturers, whereas this new arrangement will "free" them.

(A) senior
(B) swatted
(C) restrain
(D) productive

Q.3
Select the INCORRECTLY spelt word.

(A) Apraisal
(B) Affiliate
(C) Affliction
(D) Anxious

Q.4

Select the most appropriate meaning of the given idiom. 'Hold someone's feet to the fire'

(A) Agree upon a shared venture with full conviction
(B) Take big risks without knowing the consequences
(C) Think over a new proposal
(D) Insist that someone fulfil their promises or obligations

Q.5

Select the most appropriate idiom that can complete the given sentence. Recent research says that most of our cottage industries lack finance. Thus, they are _____.

(A) thrown out of gear
(B) worn and torn
(C) taking somebody for a ride
(D) under a cloud

Q.7

Select the option that rectifies the spelling of the underlined word in the given sentence. A hospital's “hygine” is the standard against which cleanliness is measured.

(A) hygeine
(B) higenie
(C) hygene
(D) hygiene

Q.8

Select the most appropriate option to fill in the blank. Active community involvement _____ a sense of belonging and contributes to the overall well-being of society.

(A) fails
(B) collects
(C) forgets
(D) fosters

Q.9

Select the correct spelling of the underlined word in the given sentence. The miracle of “resurection” is not medically proven.

(A) resurrection
(B) ressurrection
(C) ressurection
(D) resurrecsion

Q.10

Parts of the following sentence have been given as options. Select the option that contains an error. The team with a high motivation level succeeded in its project despite of the challenges it faced.

(A) despite of the challenges it faced.
(B) The team with
(C) a high motivation level
(D) succeeded in its project

Q.11

The given sentence has an error. Select the option that correctly rectifies the error. India is not on war with Pakistan.

(A) not of
(B) not at
(C) at Pakistan
(D) for Pakistan

Q.12

Select the most appropriate ANTONYM of the underlined word in the given sentence. The “mellow” tune of the flute soothes me.

(A) lyrical
(B) harsh
(C) quiet
(D) golden

Q.13

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. This boy performed / extremely good / in the / dance competition.

(A) in the
(B) This boy performed
(C) extremely good
(D) dance competition

Q.14

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. It is “easiest” to stay calm than to protest.

(A) the easier
(B) easy
(C) more easier
(D) easier

Q.15

Select the appropriate option to fill in the blank. I haven't seen him _____ last week.

(A) beyond
(B) for
(C) to
(D) since

Q.16

Select the most appropriate ANTONYM of the given word. Disposed

(A) Detained
(B) Minded
(C) Pretended
(D) Retained

Q.17

Select the most appropriate option to fill in the blank. Some types of _____ suck blood.

(A) leak
(B) leach
(C) leech
(D) latch

Q.18

Parts of the following sentence have been given as options. Select the option that contains an error. Nagamani, a brave student, lodged a F.I.R. against the criminal.

(A) student, lodged
(B) against the criminal
(C) a F.I.R.
(D) Nagamani, a brave

Q.19

Select the most appropriate ANTONYM of the given word. Sloppy

(A) Dashing
(B) Untidy
(C) Rough
(D) Humid

Q.20

The following sentence has been divided into four segments. Identify the segment that has a grammatical error. He will have to / knuckle up really seriously, / if he wants to / crack the upcoming deal.

(A) if he wants to
(B) knuckle up really seriously
(C) crack the upcoming deal.
(D) He will have to

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Throughout human history, there have been numerous, strange and unexplainable events that have occurred on our (1)_____. From mysterious disappearances to bizarre natural phenomena, these (2)_____ continue to captivate our imagination and spark curiosity. On 30 June 1908, a massive explosion (3)_____ in the remote Tunguska region of Siberia. The (4)_____ of the explosion is still debated, but it is believed to have been caused by a meteor or comet. Another fascinating phenomenon, the Bermuda Triangle is a region in the western part of the North Atlantic Ocean where ships and planes have mysteriously (5)_____.

Select the most appropriate option to fill in blank number 1.

(A) planet
(B) home
(C) neighbourhood
(D) country

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Throughout human history, there have been numerous, strange and unexplainable events that have occurred on our (1)_____. From mysterious disappearances to bizarre natural phenomena, these (2)_____ continue to captivate our imagination and spark curiosity. On 30 June 1908, a massive explosion (3)_____ in the remote Tunguska region of Siberia. The (4)_____ of the explosion is still debated, but it is believed to have been caused by a meteor or comet. Another fascinating phenomenon, the Bermuda Triangle is a region in the western part of the North Atlantic Ocean where ships and planes have mysteriously (5)_____.

Select the most appropriate option to fill in blank number 2.

(A) adventures
(B) events
(C) changes
(D) advent

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Throughout human history, there have been numerous, strange and unexplainable events that have occurred on our (1)_____. From mysterious disappearances to bizarre natural phenomena, these (2)_____ continue to captivate our imagination and spark curiosity. On 30 June 1908, a massive explosion (3)_____ in the remote Tunguska region of Siberia. The (4)_____ of the explosion is still debated, but it is believed to have been caused by a meteor or comet. Another fascinating phenomenon, the Bermuda Triangle is a region in the western part of the North Atlantic Ocean where ships and planes have mysteriously (5)_____.

Select the most appropriate option to fill in blank number 3.

(A) conspired
(B) appeared
(C) took place
(D) came about

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Throughout human history, there have been numerous, strange and unexplainable events that have occurred on our (1)_____. From mysterious disappearances to bizarre natural phenomena, these (2)_____ continue to captivate our imagination and spark curiosity. On 30 June 1908, a massive explosion (3)_____ in the remote Tunguska region of Siberia. The (4)_____ of the explosion is still debated, but it is believed to have been caused by a meteor or comet. Another fascinating phenomenon, the Bermuda Triangle is a region in the western part of the North Atlantic Ocean where ships and planes have mysteriously (5)_____.

Select the most appropriate option to fill in blank number 4.

(A) commencement
(B) beginning
(C) cause
(D) stages

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Throughout human history, there have been numerous, strange and unexplainable events that have occurred on our (1)_____. From mysterious disappearances to bizarre natural phenomena, these (2)_____ continue to captivate our imagination and spark curiosity. On 30 June 1908, a massive explosion (3)_____ in the remote Tunguska region of Siberia. The (4)_____ of the explosion is still debated, but it is believed to have been caused by a meteor or comet. Another fascinating phenomenon, the Bermuda Triangle is a region in the western part of the North Atlantic Ocean where ships and planes have mysteriously (5)_____.

Select the most appropriate option to fill in blank number 5.

(A) concealed
(B) invisible
(C) disappeared
(D) besotted

Answer key

Numerical and Mathematical Ability

Q.3 **(D)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(B)**	Q.7 **(A)**	Q.8 **(D)**
Q.10 **(C)**	Q.11 **(D)**	Q.13 **(D)**	Q.15 **(D)**	Q.16 **(B)**	Q.17 **(B)**
Q.18 **(C)**	Q.19 **(A)**	Q.20 **(A)**			

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(A)**	Q.7 **(B)**	Q.8 **(C)**
Q.9 **(A)**	Q.12 **(A)**	Q.13 **(B)**	Q.14 **(D)**	Q.15 **(A)**	Q.16 **(B)**
Q.17 **(D)**	Q.18 **(A)**	Q.20 **(D)**			

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(C)**
Q.7 **(B)**	Q.8 **(D)**	Q.9 **(B)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(B)**	Q.17 **(A)**	Q.18 **(D)**
Q.19 **(D)**	Q.20 **(B)**	Q.21 **(C)**	Q.22 **(A)**	Q.23 **(C)**	Q.24 **(D)**
Q.25 **(A)**					

English Language and Comprehension

Q.2 **(C)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(A)**	Q.7 **(D)**	Q.8 **(D)**
Q.9 **(A)**	Q.10 **(A)**	Q.11 **(B)**	Q.12 **(B)**	Q.13 **(C)**	Q.14 **(D)**
Q.15 **(D)**	Q.16 **(D)**	Q.17 **(C)**	Q.18 **(C)**	Q.19 **(A)**	Q.20 **(B)**
Q.21 **(A)**	Q.22 **(B)**	Q.23 **(C)**	Q.24 **(C)**	Q.25 **(C)**	

Explanations

Numerical and Mathematical Ability

Q.3 · Answer (D) 270

SI = 3000 × 3 × 3/100 = ₹270.

Q.4 · Answer (C) 72

$8 = 2^3$, $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$. The LCM takes the highest powers: $2^3 \times 3^2 = 72$.

Q.5 · Answer (A) 300

SI = 2500 × 3 × 4/100 = ₹300.

Q.6 · Answer (B) 102

A single 35% discount takes off ₹1,050. The successive discounts leave 3,000 × 0.8 × 0.9 × 0.95 = ₹2,052, taking off ₹948. The difference is ₹102.

Q.7 · Answer (A) 15 km/h

From 10:00 a.m. to 2:00 p.m. is 4 hours. The cab took 20/60 = 1/3 hour and the train 240/80 = 3 hours, leaving 2/3 hour for the remaining 270 − 260 = 10 km: 10 ÷ 2/3 = 15 km/h.

Q.8 · Answer (D) 20,000

Shikha's cost × 0.95 × 1.10 × 0.80 = 16,720, so her cost = 16,720/0.836 = ₹20,000.

Q.10 · Answer (C) 40 years

Three years ago the three ages totalled 3 × 27 = 81, so now they total 81 + 9 = 90. Five years ago the wife and child totalled 2 × 20 = 40, so now 50. The husband is 90 − 50 = 40 years old.

Q.11 · Answer (D) ₹6,500

CP = 7,930/1.22 = ₹6,500.

Q.13 · Answer (D) ₹75

(15 × 80 + 5 × 60)/20 = (1,200 + 300)/20 = ₹75 a kg.

Q.15 · Answer (D) 25 years

The 39 students total 39 × 15 = 585 years. With the teacher the 40 average 15 years 3 months (15.25 years), a total of 610, so the teacher is 610 − 585 = 25.

Q.16 · Answer (B) 13 years

The 40 students total 40 × 12 = 480 years. Afterwards 60 students average 12 years 4 months (37/3 years), a total of 740. The 20 newcomers total 260 years: an average of 13 years.

Q.17 · Answer (B) 32.16%

1.12 × 1.18 = 1.3216: a single increase of 32.16%.

Q.18 · Answer (C) 240 seconds

Running the same way, the faster gains 15 − 12 = 3 km/h = 5/6 m/s. They meet again when he has gained one lap of 200 m: 200 ÷ 5/6 = 240 seconds.

Q.19 · Answer (A) 1012

Length = 68 × 11/34 = 22 m and breadth = 68 × 23/34 = 46 m, so the area is 22 × 46 = 1012 m^2.

Q.20 · Answer (A) 24.5

p = kq with k = 7/6, so when q = 21, p = 7/6 × 21 = 24.5.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) ILQRP

Each letter moves 3 places back: B → Y, E → B, A → X, R → O. So L → I, O → L, T → Q, U → R and S → P: ILQRP.

Q.2 · Answer (B) 7
After the interchanges: 28 + 18 ÷ 6 – 6 × 4 = 28 + 3 – 24 = 7.

Q.3 · Answer (D) 8
Each term is the one before divided by 5: 40 ÷ 5 = 8.

Q.4 · Answer (A) 81
The second number is five times the first plus 1: 12 × 5 + 1 = 61 and 15 × 5 + 1 = 76. Likewise 16 × 5 + 1 = 81.

Q.7 · Answer (B) rt
'smart' is the only word in both sentences and 'rt' the only code in both, so smart = rt.

Q.8 · Answer (C) EIQ
Every letter moves 6 places forward each time, wrapping round after Z: G, M, S, Y, E; K, Q, W, C, I; S, Y, E, K, Q. The next cluster is EIQ.

Q.9 · Answer (A) (15, 127, 7)
In each set the middle number is the product of the other two plus their sum: 12 × 5 + 12 + 5 = 77 and 21 × 3 + 21 + 3 = 87. Likewise 15 × 7 + 15 + 7 = 127.

Q.12 · Answer (A) One
Taking the seats anticlockwise, the direction of each person's right: E, S, G, F, R, D, T. G is second to E's right, one person sits between G and R and one between R and T, D is next to T and F is second to D's left. Going left from S, only E sits before T: one person.

Q.13 · Answer (B) – and +
Interchanging – and +: 45 + 33 ÷ 3 × 4 – 67 = 45 + 44 – 67 = 22.

Q.14 · Answer (D) WJIUOYVM
The letters move +4 and –6 alternately, wrapping round the alphabet: J → N, O → I, K → O, E → Y. So S → W, P → J, E → I, A → U, K → O, E → Y, R → V and S → M: WJIUOYVM.

Q.15 · Answer (A) Only conclusion (II) follows.
All crocs are frogs, so some frogs are crocs (II). Crocs are both frogs and turtles, but nothing says every turtle is a frog, so I does not follow.

Q.16 · Answer (B) 2
'chairs' is in the first two sentences and 3 in both their codes; 'old' is in the first and third, which share 5. That leaves 2 for 'are'.

Q.17 · Answer (D) 2
Taking the seats anticlockwise, the direction of each person's right: C, B, F, D, E, A. E is second to C's left, C sits between A and B, and F is third to A's right (with A on C's right, F would land on E's seat). Going left from D, F and B sit before C: two people.

Q.18 · Answer (A) Neither conclusion [1] nor [2] follows
Every bongo is a sitar and no cymbal is a sitar, so no bongo is a cymbal and 2 is false. All bongos being sitars does not make all sitars bongos, so 1 does not follow either.

Q.20 · Answer (D) BQLA
The first two letters move 3 places back and the last two 4 forward: Q → N, F → C, R → V, G → K. So E → B, T → Q, H → L and W → A (wrapping round): BQLA.

General Awareness

Q.1 · Answer (C) Graduate and post-graduate
SSC's key gives graduates and post-graduates. In India the unemployment rate rises with education: the India Employment Report 2024 of the ILO and the Institute for Human Development put it at 29.1% for young graduates in 2022, against 3.4% for young people who could not read or write.

Q.2 · Answer (B) Kerala
Kerala launched the 'Pride' project in June 2023, run by the Kerala Knowledge Economy Mission and the Social Justice Department, to find jobs for transgender people.

Q.3 · Answer (C) vimana
In South Indian (Dravidian) temples the pyramidal tower over the sanctum is the vimana; in North Indian (Nagara) temples it is the shikhara.

Q.4 · Answer (A) Investigating and reporting on financial transactions
The CAG audits the accounts of the Union and the states and reports on them to the President and the Governors, who lay the reports before the legislatures (Article 151).

Q.5 · Answer (B) Tamil Nadu
Coimbatore in Tamil Nadu, called the 'Manchester of South India', is a centre of the cotton textile industry.

Q.6 · Answer (C) Ladakh
NCERT's division of India by rainfall puts Ladakh, with most of western Rajasthan and parts of the Peninsula, among the 'Areas of Inadequate Rainfall', with less than 50 cm a year. Haryana is among the 'Areas of Low Rainfall' (50 to 100 cm), and Meghalaya and the Andaman and Nicobar Islands get heavy rain.

Q.7 · Answer (B) Vijayanagara
Hampi Utsava, held at Hampi in Karnataka, recreates the splendour of the Vijayanagara Empire, whose capital Hampi was.

Q.8 · Answer (D) Starch, alginates, pectins and chitosans
Starch, alginates, pectins and chitosans are all edible polysaccharides. Each of the other options includes something that is not a polysaccharide: actin and myosin are proteins, glucose is a monosaccharide and sucrose a disaccharide.

Q.9 · Answer (B) Rajesh Gopinathan
Rajesh Gopinathan, TCS's CEO and MD until 2023, joined IIT Bombay in December 2023. N. Chandrasekaran, his predecessor at TCS, chairs Tata Sons.

Q.10 · Answer (C) CO_2
Carbon dioxide is made of molecules held together by covalent bonds. NaCl, $MgSO_4$ and CaO are ionic compounds, made of ions rather than molecules.

Q.11 · Answer (B) Phosphorus and nitrogen
Excess nitrogen and phosphorus, from fertiliser run-off and sewage, feed the algal blooms.

Q.12 · Answer (D) Andhra Pradesh
Mangalampalli Balamuralikrishna was born in 1930 at Sankaraguptam in the East Godavari district of Andhra Pradesh.

Q.13 · Answer (D) Amitav Ghosh
'The Hungry Tide' (2004), set in the Sundarbans, is by Amitav Ghosh.

Q.14 · Answer (A) 79 to 122
Articles 79 to 122, Chapter II of Part V, deal with Parliament.

Q.15 · Answer (D) UAE
The 2023 ACC Under-19 Asia Cup was held in the UAE in December 2023; Bangladesh beat the hosts in the final at Dubai.

Q.16 · Answer (B) Womesh Chandra Banerjee
Womesh Chandra Banerjee (W. C. Bonnerjee) presided over the first session of the Indian National Congress, at Bombay in December 1885.

Q.17 · Answer (A) Na_2CO_3
Baking soda ($NaHCO_3$) is made from sodium chloride, water, carbon dioxide and ammonia: $NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$. Sodium carbonate ($Na_2CO_3$) is what baking soda gives on heating, not a raw material for it.

Q.18 · Answer (D) India
Delhi hosted the 2010 Commonwealth Games.

Q.19 · Answer (D) Karnataka
Dollu Kunitha, a vigorous drum dance, is a folk dance of Karnataka.

Q.20 · Answer (B) 60 days
Under Article 190(4), if a member of a state legislature is absent from all its meetings for 60 days without its permission, the House may declare the seat vacant.

Q.21 · Answer (C) Niranjan Rout
The Odissi guru Niranjan Rout was chosen for the Sangeet Natak Akademi Award for 2022.

Q.22 · Answer (A) Shashanka
Shashanka ruled Gauda (Bengal) in the early 7th century and killed Harsha's brother Rajyavardhana. Grahavarman was the Maukhari king of Kannauj and Rudradaman a Shaka ruler of western India.

Q.23 · Answer (C) Cotton clothes made in England had driven Indian clothes out of the markets of Europe and America in the beginning of the 20th century.
English cotton goods drove Indian textiles out of their markets in Europe, America and Africa by the beginning of the 19th century, not the 20th. The other three statements are correct.

Q.24 · Answer (D) Bihar
In the 2011 census about 89% of Bihar's people were rural, against about 86% in Assam, 78% in Uttar Pradesh and 68% in West Bengal.

Q.25 · Answer (A) Mount Abu
The Dilwara temples stand at Mount Abu in Rajasthan; the earliest was built in 1031 by Vimal Shah, a minister of the Solanki king Bhima I.

English Language and Comprehension

Q.2 · Answer (C) restrain
To 'free' is to release; its opposite is 'restrain'.

Q.3 · Answer (A) Apraisal
The correct spelling is 'appraisal'.

Q.4 · Answer (D) Insist that someone fulfil their promises or obligations
To hold someone's feet to the fire is to press them to keep their promises or do their duty.

Q.5 · Answer (A) thrown out of gear
'Thrown out of gear' means put out of working order: without finance the industries cannot run smoothly.

Q.7 · Answer (D) hygiene
The correct spelling is 'hygiene'.

Q.8 · Answer (D) fosters
To 'foster' is to encourage or develop: involvement fosters a sense of belonging.

Q.9 · Answer (A) resurrection
The correct spelling is 'resurrection', with a double 'r'.

Q.10 · Answer (A) despite of the challenges it faced.
'Despite' takes no 'of': 'despite the challenges', or 'in spite of the challenges'.

Q.11 · Answer (B) not at
The phrase is 'at war': 'India is not at war with Pakistan.'

Q.12 · Answer (B) harsh
'Mellow' means soft and pleasant; its opposite is 'harsh'.

Q.13 · Answer (C) extremely good
'Performed' is a verb, so it takes the adverb: 'extremely well', not 'extremely good'.

Q.14 · Answer (D) easier
'Than' compares two things, so the comparative 'easier' is needed; 'more easier' doubles it.

Q.15 · Answer (D) since
'Since' goes with a point in time ('last week'); 'for' would need a length of time.

Q.16 · Answer (D) Retained
To dispose of something is to get rid of it; its opposite is 'retained', kept.

Q.17 · Answer (C) leech
A leech is a worm that sucks blood; to 'leach' is to drain out of soil.

Q.18 · Answer (C) a F.I.R.
'F' is read 'ef', which begins with a vowel sound, so 'an F.I.R.'

Q.19 · Answer (A) Dashing
'Sloppy' means careless and untidy; 'dashing' means smart and stylish. 'Untidy' is a synonym.

Q.20 · Answer (B) knuckle up really seriously
The phrasal verb is 'knuckle down', to start working seriously; 'knuckle up' does not have this meaning.

Q.21 · Answer (A) planet
The events have occurred on 'our planet', the Earth.

Q.22 · Answer (B) events
'These events' points back to the 'strange and unexplainable events' of the first sentence.

Q.23 · Answer (C) took place
An explosion 'took place', that is happened, at a particular time and place.

Q.24 · Answer (C) cause
The 'cause' of the explosion is debated: it is believed to have been caused by a meteor or comet.

Q.25 · Answer (C) disappeared
Ships and planes have mysteriously 'disappeared' in the Bermuda Triangle, echoing the 'mysterious disappearances' above.

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